

SANTA MARIA DI LEUCA, Italy

WMO: 163600

Lat: 39.811N

Lon: 18.342E

Elev: 367

StdP: 14.50

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.7	40.5	22.3	17.0	43.5	26.0	20.3	45.0	30.5	52.1	27.8	51.7	13.2	20	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	89.8	71.6	87.6	72.0	85.7	71.5	78.9	83.5	77.4	82.3	76.0	81.3	9.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	145.1	81.8	75.9	137.3	80.9	74.4	130.2	79.9	42.7	83.5	41.1	82.4	39.8	81.4	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.7	20.2	33.3	94.8	3.4	3.7	30.9	97.4	28.9	99.6	27.0	101.6	24.5	104.3		
			DB	WB	30.1	81.3	2.8	2.2	28.1	82.8	26.4	84.1	24.9	85.3	22.8	86.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	50.1	50.6	53.5	58.1	65.3	73.2	78.8	79.1	72.4	65.6	58.9	52.6
	DBStd	11.23	4.82	4.48	4.20	4.01	4.54	4.95	3.89	3.76	4.33	4.30	5.08	5.01
	HDD50	157	57	43	18	2	0	0	0	0	0	0	5	32
	HDD65	2105	462	403	356	209	51	2	0	0	2	45	189	386
	CDD50	4999	60	61	127	246	476	697	892	903	671	483	272	111
	CDD65	1470	0	0	0	3	61	249	427	438	223	63	6	0
	CDH74	10392	0	0	0	3	134	1420	3837	3995	962	41	0	0
	CDH80	2701	0	0	0	0	6	308	1139	1136	110	2	0	0
Wind	WSAvg	10.3	11.2	11.9	10.9	9.9	9.2	9.2	9.5	9.4	9.7	10.2	10.9	11.5
Precipitation	PrecAvg	26.1	3.0	2.4	2.8	1.5	1.0	0.7	0.7	0.9	1.9	4.0	4.1	3.8
	PrecMax	43.9	11.4	6.9	8.9	4.8	3.4	3.1	4.4	4.2	5.0	15.2	12.7	9.4
	PrecMin	9.1	0.1	0.2	0.4	0.2	0.0	0.0	0.0	0.0	0.2	0.2	0.4	1.1
	PrecStd	8.2	2.3	1.8	2.0	1.0	0.7	0.8	1.0	1.2	1.5	3.1	3.5	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	61.2	65.0	73.0	80.7	89.6	93.3	93.7	86.0	77.5	70.8	63.0
		MCWB	55.7	54.6	55.6	60.1	65.4	68.4	70.7	72.7	69.4	69.4	66.1	59.3
	2%	DB	58.8	59.1	62.4	69.1	77.4	86.3	90.0	90.0	82.8	74.9	68.6	61.8
		MCWB	55.0	54.5	55.3	58.4	63.9	70.3	71.0	73.2	70.8	68.6	64.7	58.3
	5%	DB	57.5	57.6	60.4	66.2	75.1	83.6	87.7	87.7	80.7	73.2	67.2	60.6
		MCWB	54.2	54.0	54.9	56.9	64.6	69.7	71.5	73.0	71.7	67.3	63.4	57.1
	10%	DB	56.5	56.7	58.9	63.9	72.5	81.0	85.7	85.7	78.9	71.7	65.6	59.2
		MCWB	53.4	53.3	54.7	56.8	63.5	69.1	71.4	72.1	71.8	66.6	61.7	55.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.3	57.5	59.6	63.9	71.0	78.2	80.1	82.2	78.0	72.7	67.6	60.6
		MCDB	59.0	58.9	62.2	68.6	75.0	83.2	84.5	85.8	80.2	75.7	69.7	62.3
	2%	WB	55.9	56.0	57.6	61.2	68.3	75.9	78.1	79.7	76.2	70.6	65.6	59.0
		MCDB	57.8	57.9	60.2	65.3	73.4	81.0	83.4	84.1	79.2	73.5	68.0	61.0
	5%	WB	55.0	54.9	56.3	59.7	66.7	73.9	76.7	78.1	74.7	69.0	64.0	57.6
		MCDB	57.0	57.0	58.9	63.3	71.7	79.4	82.3	82.9	78.4	72.0	66.7	59.9
	10%	WB	53.8	53.8	55.2	58.5	65.2	72.1	75.4	76.6	73.0	67.6	62.3	56.5
		MCDB	56.1	56.1	57.8	61.8	70.3	77.8	81.5	81.8	77.3	70.7	65.1	59.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.2	7.5	8.1	8.8	9.6	10.5	10.3	9.0	8.1	7.3	6.9
		MCDBR	6.8	7.7	9.2	11.8	12.1	12.9	12.8	12.9	10.1	8.8	7.4	6.7
		MCWBR	6.0	6.2	7.0	8.0	8.1	9.7	10.8	10.5	8.0	6.6	6.3	5.9
	5% WB	MCDBR	5.8	6.2	7.3	8.4	9.7	10.0	10.5	9.9	8.4	7.1	6.5	5.7
		MCWBR	5.9	6.0	6.8	7.1	7.8	9.0	10.3	9.9	7.9	7.0	6.2	5.8
Clear-Sky Solar Irradiance	taub	0.343	0.365	0.403	0.443	0.432	0.419	0.415	0.428	0.420	0.398	0.372	0.349	
	taud	2.471	2.392	2.284	2.179	2.270	2.351	2.369	2.333	2.354	2.386	2.417	2.456	
	Ebn at Noon	259	269	269	265	270	273	273	266	260	253	245	247	
	Edh at Noon	25	31	38	45	43	39	38	39	36	31	27	24	
All-Sky Solar Radiation	RadAvg	625	885	1318	1734	2113	2391	2390	2096	1578	1094	710	560	
	RadStd	54	84	114	127	87	102	82	88	83	95	54	69	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	-148	+229	N/A

Nomenclature: See separate page